Outlier Detection Report (Boxplot / IQR Method)

This report summarizes the process and results of extreme outlier detection in the dataset using the Interquartile Range (IQR) method based on boxplot analysis.

1. Data Overview
The dataset contains 76,000 observations with both categorical and numerical variables. Outlier detection focuses on numerical features: *Inventory Level, Units Sold, Units Ordered, Price, Discount, Competitor Pricing,* and *Demand*.

2. Outlier Detection Method
The boxplot method identifies outliers based on quartiles (Q1, Q3) and the Interquartile Range (IQR = Q3 – Q1).
Thresholds are defined as follows: Mild outliers: values outside [Q1 – 1.5 × IQR, Q3 + 1.5 × IQR] Extreme outliers: values outside [Q1 – 3 × IQR, Q3 + 3 × IQR] Any sample exceeding these thresholds is marked as an outlier.

3. Python Implementation
A simplified version of the detection code is as follows: import pandas as pd import matplotlib.pyplot as plt numeric_cols = ['Inventory Level', 'Units Sold', 'Units Ordered', 'Price', 'Discount', 'Competitor Pricing', 'Demand'] for col in numeric_cols: Q1 = data[col].quantile(0.25) Q3 = data[col].quantile(0.75) IQR = Q3 - Q1 extreme_lower = Q1 - 3 * IQR extreme_upper = Q3 + 3 * IQR data[f'{col}_is_extreme_outlier'] = ((data[col] < extreme_lower) | (data[col] > extreme_upper))

4. Results and Interpretation
Samples identified as extreme outliers include rows such as 6745, 7886, 15994, 16165, and 16254. These cases exhibit unusually high or low values in *Price*, *Units Ordered*, and *Demand*. Possible reasons include seasonal promotions, regional variations, or data entry errors.

5. Conclusion
The IQR-based boxplot method effectively identified extreme outliers within the dataset. Further business analysis is required to decide whether these data points represent genuine business fluctuations or anomalies that should be removed or adjusted before modeling.